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|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| 7(a)(i)   | As diameter of the wire $d$ increases, resistance of the wire $R$ decreases.                                                                                                                                                                                                                | B1                           |
| 7(a)(ii)  | Taking natural logarithm of $R = kd^n$ , <u><math>\lg R = n \lg d + \lg k</math></u> . $R = kd^n$ is valid because the graph <u><math>\lg R</math> against <math>\lg d</math> is a straight line.</u>                                                                                       | B1<br>B1                     |
| 7(b)(i)   | Plotting the point for $d = 0.46$ mm correctly.                                                                                                                                                                                                                                             | M1                           |
| 7(b)(ii)  | Drawing the best fit line correctly.                                                                                                                                                                                                                                                        | M1                           |
| 7(b)(iii) | $n = (0.34 - 1.2) / (-0.1 + 0.53)$<br>$= -2.0$                                                                                                                                                                                                                                              | M1<br>A1                     |
| 7(c)      | $\lg(0.73) = -0.137$ .<br><br>At $\lg d = -0.137$ , $\lg R = 0.41$ .<br>$R = 10^{0.41}$<br>$= 2.57 \text{ m}\Omega$<br>$= 2.57 \times 10^{-3} \Omega$                                                                                                                                       | M1<br><br>M1<br><br>A1       |
| 7(d)      | $Y = mX + c$<br>$(1.14) = -2.0(-0.5) + c$<br>$c = 0.14$<br><br>$k = 10^{0.14} = 1.38 \text{ m}\Omega \text{ mm}^2$<br><br>$k = 4\rho L / \pi$<br>$1.38 = 4\rho(1000) / \pi$<br>$\rho = 0.00108 \text{ m}\Omega \text{ mm}$ (no mark for unit)                                               | M1<br><br>M1<br><br>M1<br>A1 |
| 7(e)(i)   | Consider boiler:<br><br>$\text{Using } P = I^2 R$<br>$5.4 \times 10^6 = I^2 (6 \times 10^3)$<br>$I = 30 \text{ A}$<br><br>From graph, when $I = 30 \text{ A}$ , $d = 0.42 \text{ mm}$<br>Since minimum diameter required is $0.42 \text{ mm}$ .<br>therefore use wire <b><u>gauge 8</u></b> | M1<br><br>M1<br><br>A1       |
| 7(e)(ii)  | Lower manufacturing cost                                                                                                                                                                                                                                                                    | B1                           |
| 7(e)(iii) | For the same resistivity, thicker wires have lower resistance and as a results, there is lower power losses in them.                                                                                                                                                                        | B1<br>B1                     |
| 7(e)(iv)  | Using $V = IR$<br>$= 30 (6.6 \times 10^{-3})$<br>$= 0.198 \text{ V}$                                                                                                                                                                                                                        | M1<br><br>A1                 |